Follow-up Semi-Structured Interview Questions

Qs in red are the ones that will be asked or not asked depending on participants' answers
Qs in blue are the ones that will be asked of all participants
E stands for Experimental Group, and these questions are specific to the E group
C stands for Control Group, and these questions are specific to the C group

Part I: Follow-up on pre-survey

Multimodal features selected:
text-based chat only → Q2
more than one option → Q3

Q1: In the pre-survey, you indicated that you only use [text-based chat] in GenAI tools. Are there any reasons not to use other multimodal features?

Q2: In the pre-survey, you indicated that you have used the multimodal features: [#, #, #]. What do you primarily use these features for?

Q3:
In the pre-survey, you indicated that you [always/often] verify the accuracy of AI-generated content. In what situations do you decide to verify, and how do you verify?

In the pre-survey, you indicated that you [sometimes/seldom] verify the accuracy of AI-generated content. What factors influence whether you decide to verify or not, and when you do, how do you verify?

In the pre-survey, you indicated that you do not typically (never) verify the accuracy of AI-generated content. Are there any reasons not to verify it?

Q4: How do you feel about the role of GenAI tools in supporting learning within your field of study?

Q5: How do you think GenAI tools impact the software engineering workflow?

Part II: Follow-up on post-study

—----------Usage Perceptions—----------

Q6 (E): How do different multimodal features affect solving each SE task? (Explain for each task)

Q6 (C): How does the usage of text prompts and text file uploads affect solving each task? (Explain for each task)

Q7 (E): Which SE tasks do you find the multimodal GenAI features most and least effective/helpful for? Please explain why.

Q7 (C): Can you think of any additional multimodal features that could have helped you complete today's SE tasks more effectively? Why or why not? (Explain for each task)

Q8: Besides the tasks covered in today's study, can you think of any other software engineering tasks that could benefit from using multimodal GenAI tools? If so, which ones and how?

Q9: Similarly, what other general challenges do you think come with using multimodal GenAI for software engineering tasks?

——————————————————————————Educational Impacts——————————————————————————

Note:
Learning — the ability to recall and repeat by acquiring, memorizing, and gathering new facts and skills.
Understanding — the ability to explain and apply by interpreting information, spotting patterns, knowing the "why" and "how," and using that knowledge to solve new, real-world problems.
For example: in a classroom context, a student may memorise a formula by learning, but when they understand it, they can explain how it works or adapt it to a new problem.

SE concepts: Nielsen's Usability Heuristics, Functional Requirements, Use Case, User Story

Q10: Do you feel that interacting with the GenAI tool today helps you learn any of the SE concepts?
→ If yes, how does it help, and how does each multimodal feature contribute most to your learning?
→ If no, why doesn't it help

Q11: Do you feel that interacting with the GenAI tool today helps you understand any of the SE concepts?
→ If yes, how does it help, and how does each multimodal feature contribute most to your understanding?
→ If no, why doesn't it help

Q12 (E): How could multimodal GenAI be integrated into SE courses or projects to better support student learning and understanding?

Q12 (C): How could GenAI be integrated into SE courses or projects to better support student learning and understanding?

——————————————————————————Improvements and Suggestions——————————————————————————

Q13: Based on your experience with Google AI Studio, are there any features or capabilities you feel are currently missing or could be enhanced?

Q14: Do you have any additional comments or suggestions?